

Section 4.4 l'Hospital's Rule

$$\text{Ex } \textcircled{1} \quad \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} \quad \frac{0}{0} \stackrel{\text{l'H}}{=} \lim_{x \rightarrow 0} \frac{-\sin x}{1} = \frac{-\sin(0)}{1} = \boxed{0}$$

$$\text{Ex } \textcircled{2} \quad \lim_{x \rightarrow \infty} \frac{\ln x}{e^x} \quad \frac{\infty}{\infty} \stackrel{\text{l'H}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x} \cdot x}{e^x \cdot x} = \lim_{x \rightarrow \infty} \frac{1}{x \cdot e^x} \rightarrow \frac{1}{\infty} = \boxed{0}$$

$$\text{Ex } \textcircled{3} \quad \lim_{x \rightarrow \infty} \frac{2^x}{x^2} \quad \frac{\infty}{\infty} \stackrel{\text{l'H}}{=} \lim_{x \rightarrow \infty} \frac{2^x \cdot \ln 2}{2x} \quad \frac{\infty}{\infty}$$

$$\stackrel{\text{l'H}}{=} \lim_{x \rightarrow \infty} \frac{2^x \cdot (\ln 2)^2}{2}$$

$$= \frac{\infty}{2}$$

$$= \boxed{\infty}$$

Ex (4) $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$

$$y = x^{\frac{1}{x}}$$

$$\ln y = \ln x^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \cdot \ln x$$

$$\ln y = \frac{\ln x}{x}$$

$$\lim_{x \rightarrow \infty} \left(\ln y = \frac{\ln x}{x} \right) \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \left(\ln y \stackrel{\text{PH}}{=} \frac{\frac{1}{x}}{1} \right)$$

$$e^{\ln y} = e^0$$

$$y = 1$$

$$\therefore \lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \boxed{1}$$

what is $\infty \cdot 0$?

$$\underline{\text{Ex}} \quad (5) \quad \lim_{x \rightarrow \infty} 17x^2 \cdot \left(\frac{1}{x^2 + x} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{17x^2}{x^2 + x}$$

$$\stackrel{1^{\text{st}}}{=} \lim_{x \rightarrow \infty} \frac{34x}{2x+1} \quad \frac{\infty}{\infty}$$

$$\stackrel{1^{\text{st}}}{=} \lim_{x \rightarrow \infty} \frac{34}{2}$$

$$= 17$$